LAPORAN PRAKTIKUM ELEKTRONIK DIGITAL

TEKNIK IMPLEMENTASI



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PROGRAM STUDI D3 TEKNIK TELEKOMUNIKASI

JURUSAN TEKNIK ELEKTRO

POLITEKNIK NEGERI BANDUNG

2024-2025

KETERANGAN

1. Kelompok : 4
2. Judul Praktek : Pratikum Teknik Implementasi
3. Tanggal Praktek : 26 Maret 2025
4. Tanggal Pengumpulan Laporan : 5 April 2025
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I. MAKSUD DAN TUJUAN PERCOBAAN

1. Menganalisis implementasi rangkaian F0 dan F1 menggunakan gerbang logika NAND dan NOR.
2. Menyederhanakan rangkaian menjadi 2 IC

II. DASAR TEORI

II.1 Integrated Circuit (IC)

Integrated Circuit (IC) adalah komponen elektronika semi konduktor yang merupakan gabungan dari berbagai komponen elektronik lain yang jumlahnya ratusan hingga ribuan komponen elektronik. Sehingga disebut sebagai rangkatan terpadu Integrated Circuit (IC). Meskipun terdiri dari komponen yang jumlahnya mulai dari ratusan hingga ribuan, namun bentuk serta ukuran komponen ini terbilang kecil. Komponen ini terbuat dari bahan semi konduktor, dimana bahan yang paling sering digunakan untuk membuat komponen ini adalah silikon. [1]

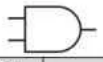
II.2 Gerbang Dasar Logika

Gerbang logika dapat didefinisikan sebagai peralatan yang dapat menghasilkan suatu output hanya bila telah ditentukan sebelumnya kondisi input yang ada. Dalam hal ini digunakan istilah gerbang karena menunjukkan keadaan terbuka atau tertutup. Terdapat beberapa gerbang logika yang digunakan secara umum. [2]

1. Gerbang AND [7408]
2. Gerbang OR [7432]
3. Gerbang NOT [7404]
4. Gerbang NAND (NOT-AND) [7400]
5. Gerbang NOR (NOT-OR) [7402]
6. Gerbang Eksklusif OR (XOR) [7486]
7. Gerbang Eksklusif NOR (NOT-OR)

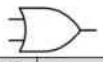
II.3 Jenis-jenis Gerbang Dasar Logika

AND



INPUT		OUTPUT
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

OR



INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

XOR



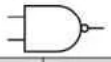
INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

NOT



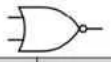
INPUT	OUTPUT
A	
0	1
1	0

NAND



INPUT		OUTPUT
A	B	
0	0	1
1	0	1
0	1	1
1	1	0

NOR



INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	0

XNOR



INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	1

II.4 Resistor

Resistor atau Tahanan adalah komponen elektronika yang berfungsi untuk mengatur kuat arus yang mengalir. Lambang untuk Resistor dengan huruf R, nilainya dinyatakan dengan cincin-cincin berwarna dalam OHM (Ω). pada teknik listrik dan elektronika terdapat dua macam resistor yang sering digunakan yaitu resistor tetap dan resistor variable. [4]

II.5 Power Supply

Power supply merupakan perangkat keras yang mampu menyuplai tenaga atau tegangan listrik secara langsung dari sumber tegangan listrik PLN ke tegangan listrik yang lainnya. Power supply memiliki input dari tegangan yang berarus alternating current (AC) dan mengubahnya menjadi arus directcurrent (DC). [5]

II.6 Jumper

Kabel jumper merupakan kabel elektrik yang mempunyai pin konektor di setiap ujungnya dan memungkinkan untuk menghubungkan dua komponen yang melibatkan Arduino tanpa memerlukan solder. Intinya, kegunaan *kabel jumper* ini digunakan sebagai konduktor listrik untuk menyambungkan rangkaian listrik. [6]

III. ALAT-ALAT YANG DIPERGUNAKAN

1. Project board
2. Multimeter Digital
3. Jumper
4. LED
5. Resistor
6. IC (7402, 7400)
7. Power supply

IV. LANGKAH KERJA DAN RANGKAIAN PERCOBAAN

1. Siapkan alat dan bahan praktikum
2. Buatlah Rangkaian
3. Rancang diagram rangkaian gerbang logika NAND dan NOR berdasarkan skema yang telah disediakan.
4. Pastikan setiap koneksi pada rangkaian sesuai dengan teori yang dipelajari.
5. Susun komponen pada breadboard sesuai dengan rancangan yang telah dibuat.
6. Hubungkan input dan output sesuai dengan konfigurasi yang diinginkan.
7. Periksa kembali koneksi untuk menghindari kesalahan pemasangan.
8. Nyalakan sumber daya dan amati respons rangkaian terhadap berbagai kombinasi input.
9. Catat keluaran yang dihasilkan oleh rangkaian NAND dan NOR.
10. Bandingkan hasil percobaan dengan tabel kebenaran yang sesuai.

V. TABEL PERCOBAAN DAN FOTO PERCOBAAN

A. Tabel Percobaan

1. F0 = $\bar{B}\bar{D} + \bar{C}D$

A	B	C	D	F	Fv	
					NAND	NOR
0	0	0	0	1	4,64 V	4, 64 V
0	0	0	1	1	4,64 V	4,64 V
0	0	1	0	1	4,64 V	4,64 V
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	0	1	1	4,64 V	4,64 V
0	1	1	0	0	0	0
0	1	1	1	0	0	0
0	0	0	0	1	4,64 V	4,64 V
0	0	0	1	1	4,64 V	4,64 V
0	0	1	0	1	4,64 V	4,64 V
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	0	1	1	4,64 V	4,64 V
0	1	1	0	0	0	0
0	1	1	1	0	0	0

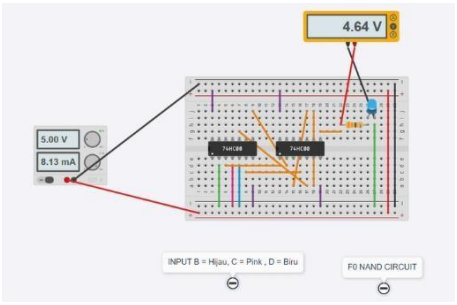
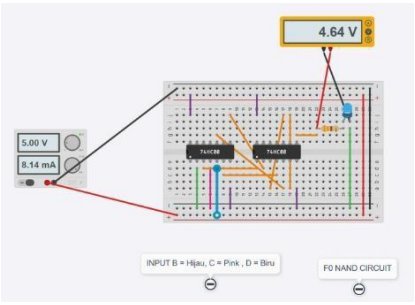
2. F1 = $\bar{A}D + A\bar{D} + A\bar{B}$

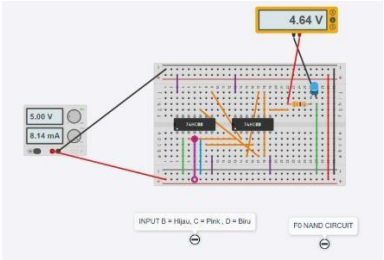
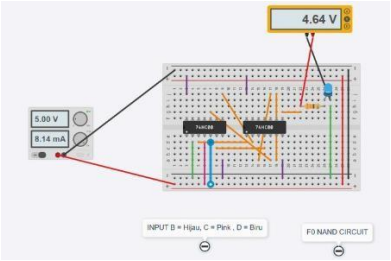
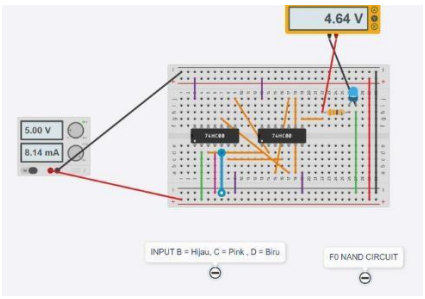
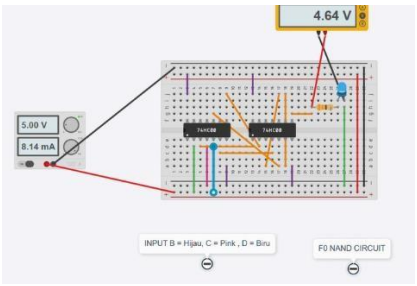
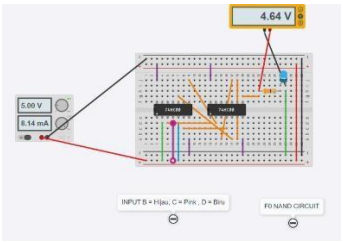
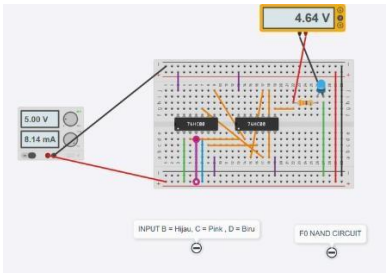
A	B	C	D	F	Fv	
					NAND	NOR
0	0	0	0	0	0	0
0	0	0	1	1	4,64 V	4,87 V
0	0	1	0	0	0	0
0	0	1	1	1	4,64 V	4,87 V

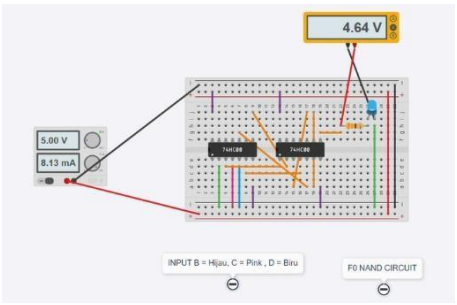
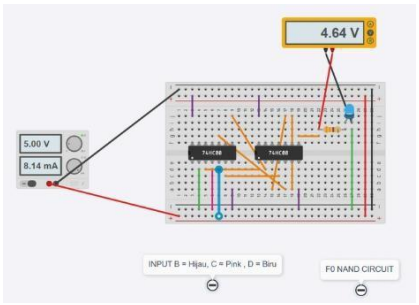
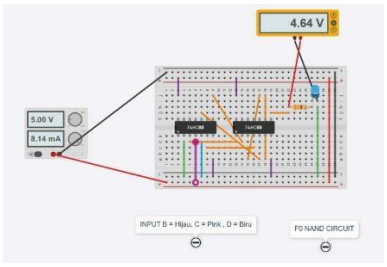
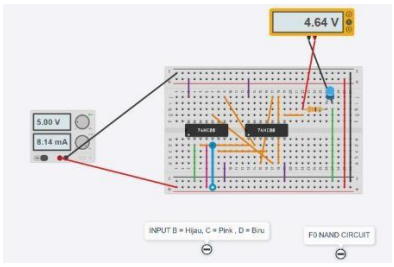
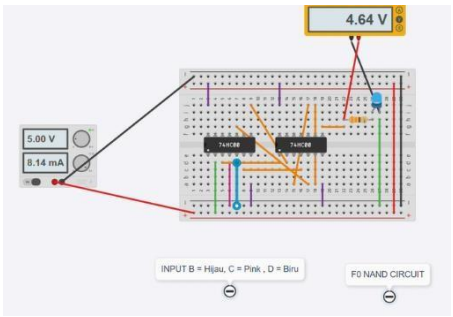
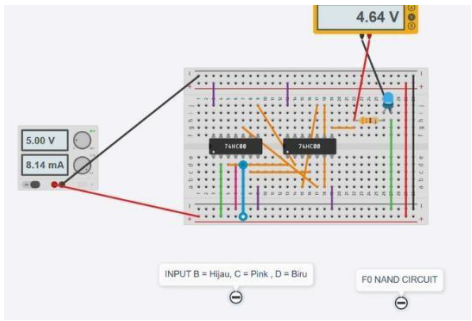
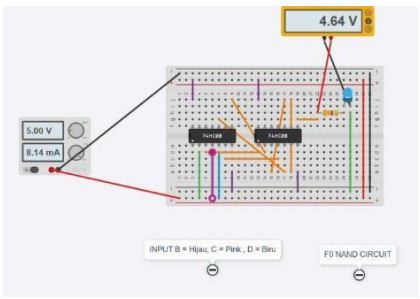
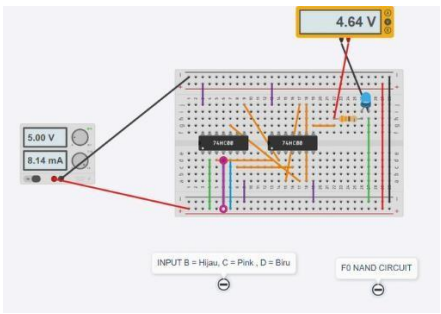
0	1	0	0	0	0	0
0	1	0	1	1	4,64 V	4,87 V
0	1	1	0	0	0	0
0	1	1	1	1	4,64 V	4,87 V
0	0	0	0	1	4,64 V	4,87 V
0	0	0	1	1	4,64 V	4,87 V
0	0	1	0	d	4,64 V	4,87 V
0	0	1	1	d	4,64 V	4,87 V
0	1	0	0	1	4,64 V	4,87 V
0	1	0	1	0	0	0
0	1	1	0	1	4,64 V	4,87 V
0	1	1	1	0	0	0

B. Foto Percobaan

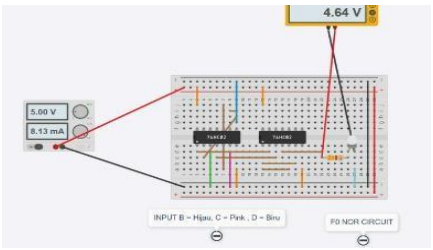
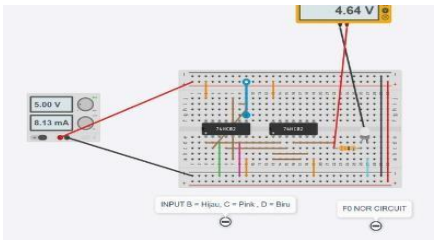
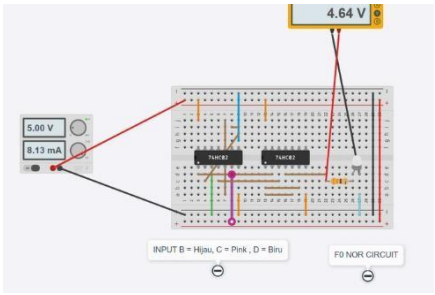
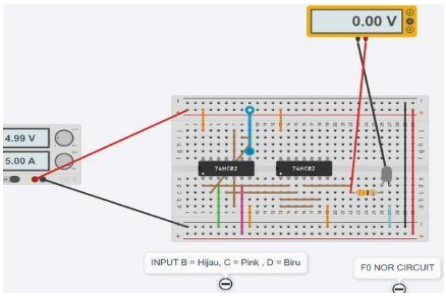
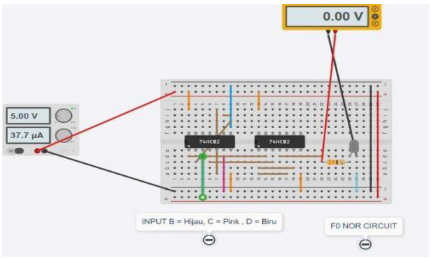
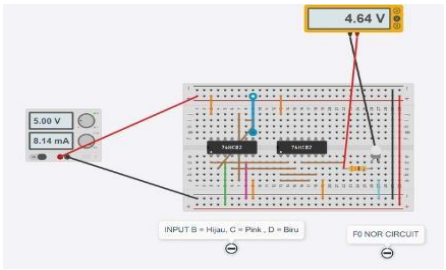
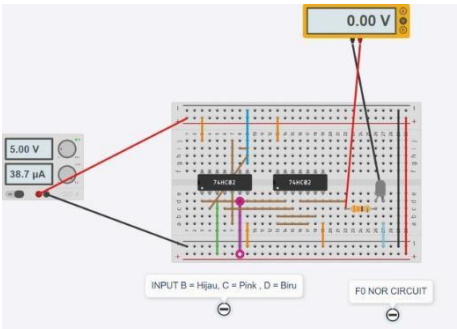
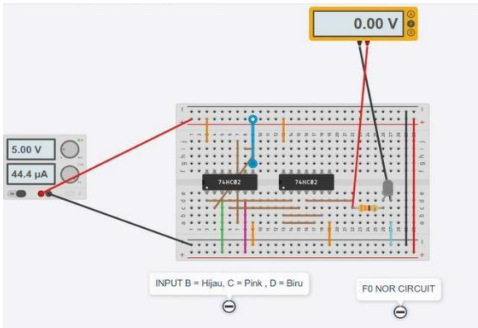
1. F0 NAND

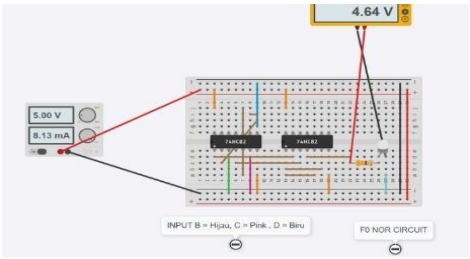
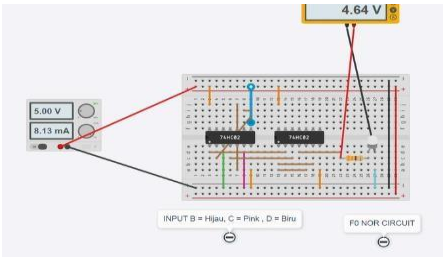
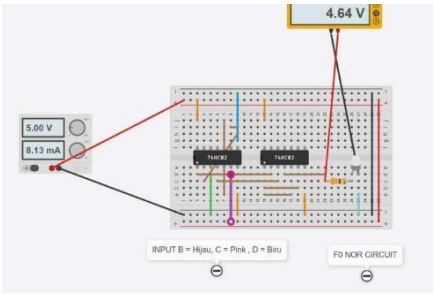
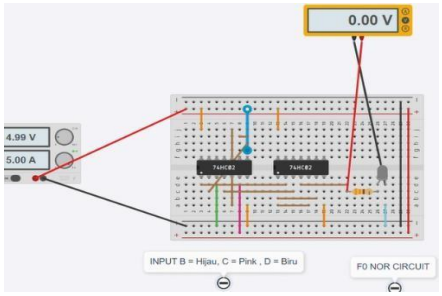
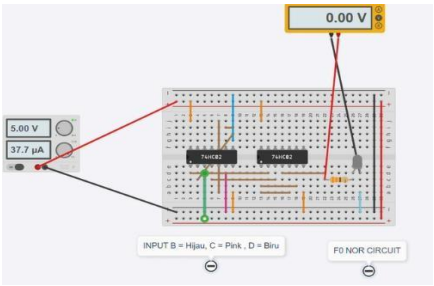
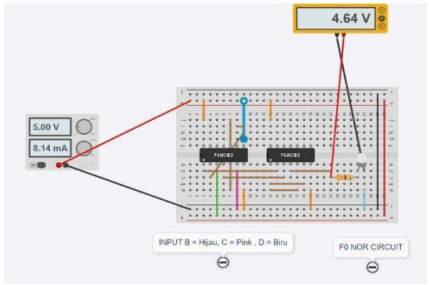
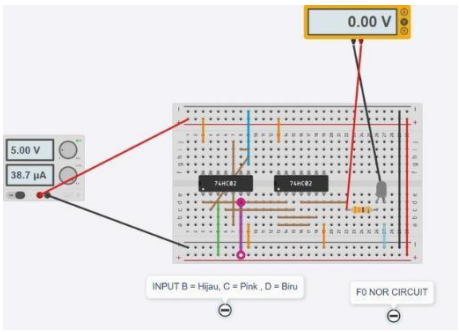
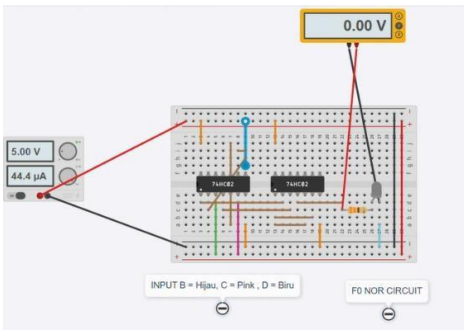
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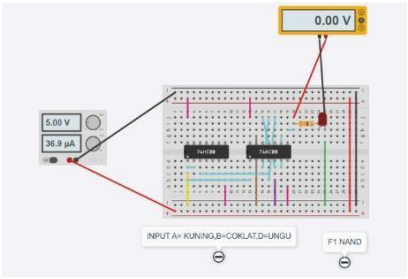
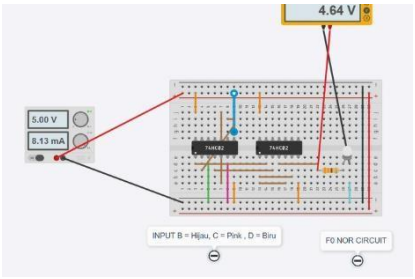
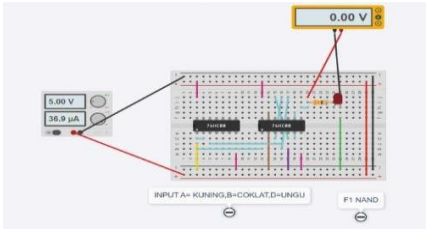
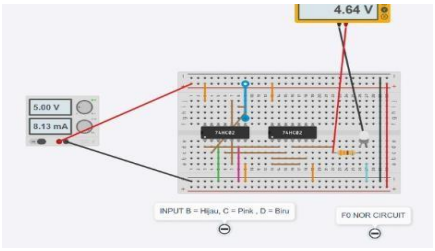
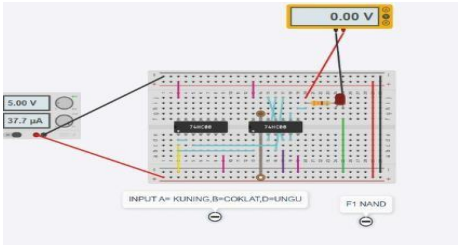
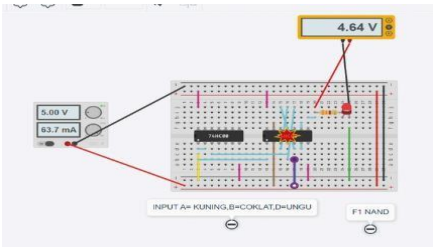
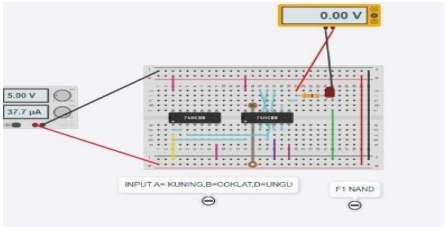
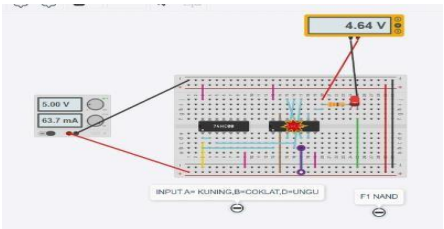
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1010 	**1011** 
1100 	**1101** 
1110 	**1111** 

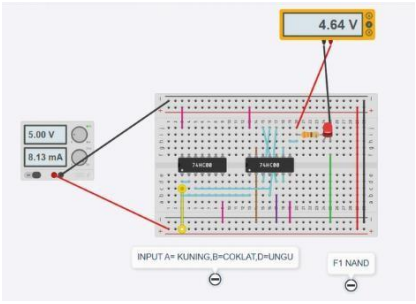
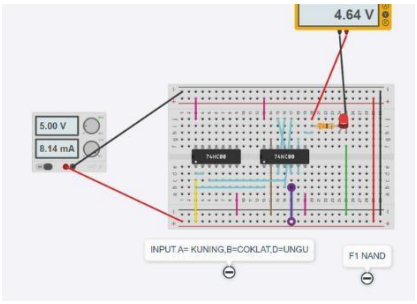
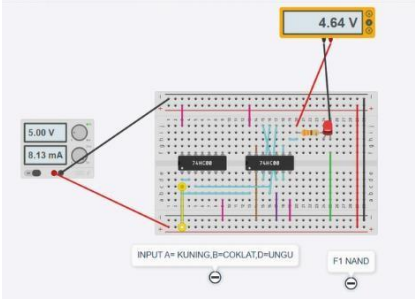
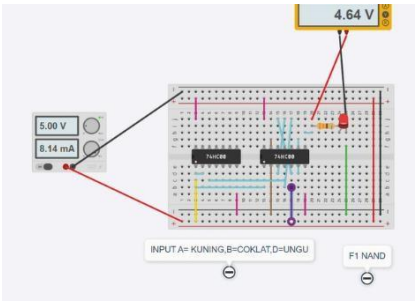
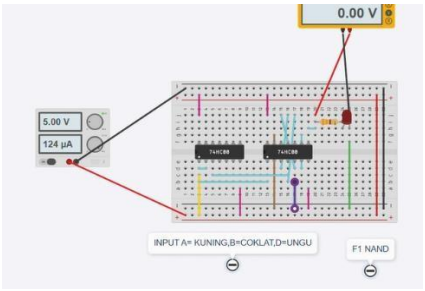
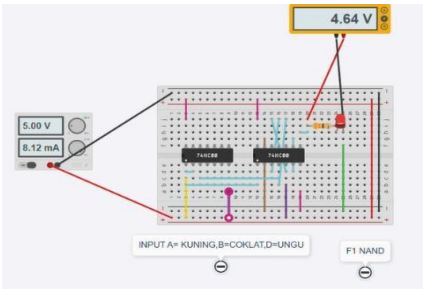
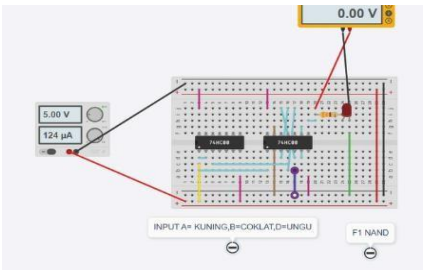
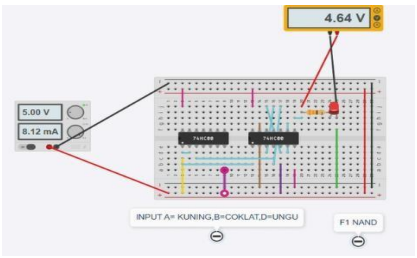
2. F0 N0R

0000	**0001**
	
0010	**0011**
	
0100	**0101**
	
0110	**0111**
	

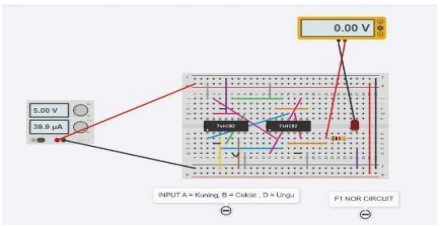
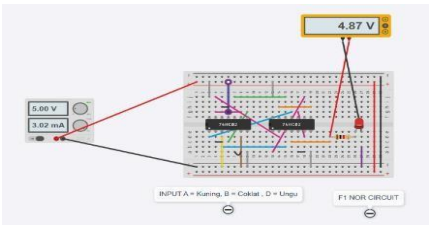
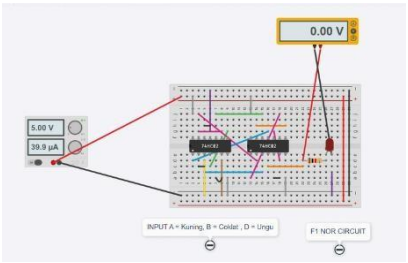
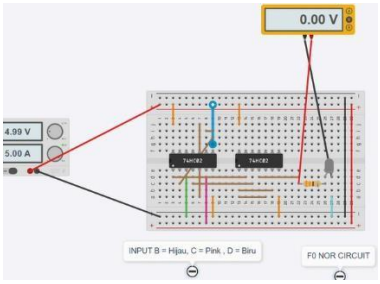
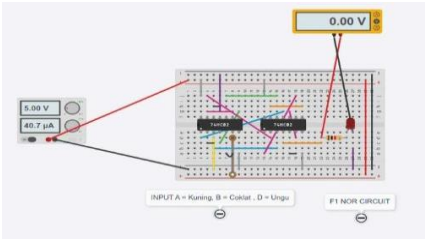
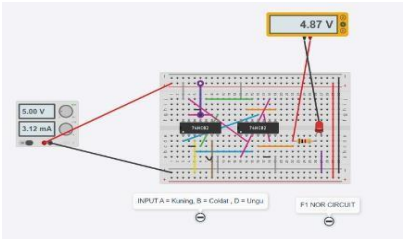
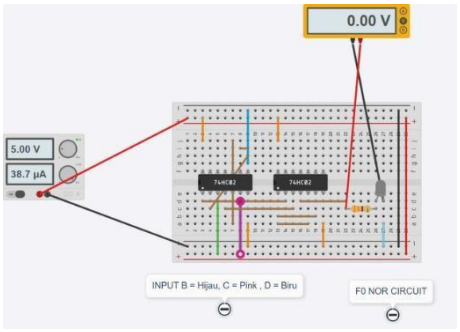
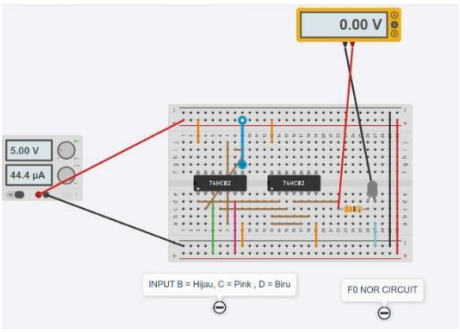
1000 	**1001** 
1010 	**1011** 
1100 	**1101** 
1110 	**1111** 

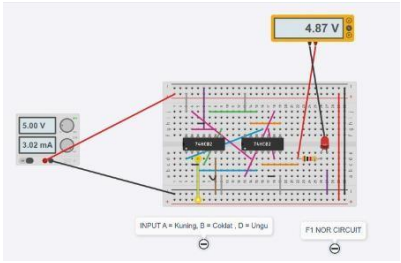
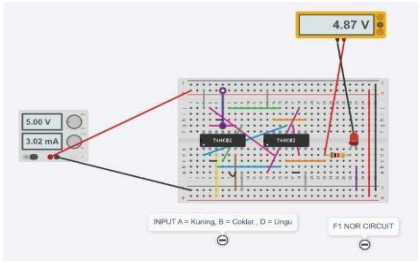
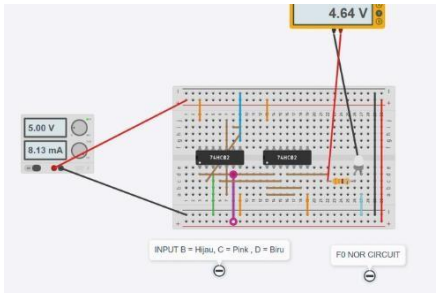
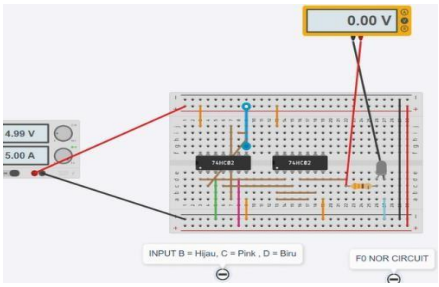
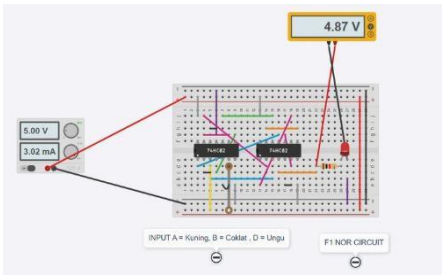
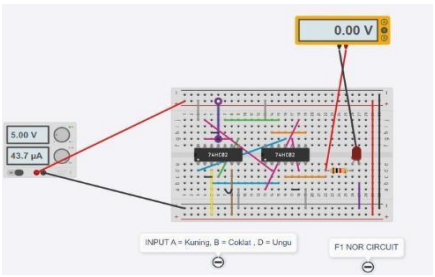
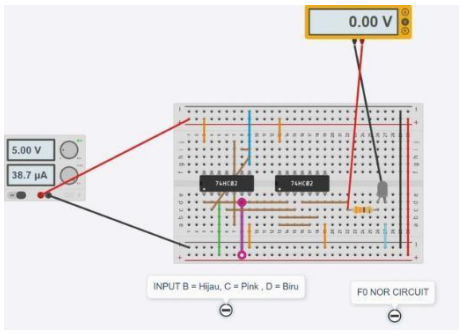
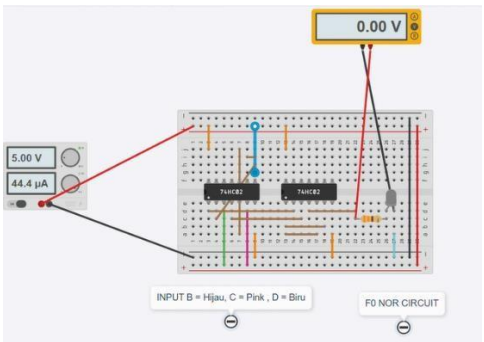
3. F1 NAND

0000	**0001**
	
0010	**0011**
	
0100	**0101**
	
0110	**0111**
	

1000 	**1001** 
1010 	**1011** 
1100 	**1101** 
1110 	**1111** 

4. F1 NOR

0000	0001
	
0010	**0011**
	
0100	**0101**
	
0110	**0111**
	

1000 	**1001** 
1010 	**1011** 
1100 	**1101** 
1110 	**1111** 

VI. ANALISA DATA

- Dari hasil pengujian rangkaian NAND dan NOR sesuai dengan tabel kebenaran
- Gerbang logika NAND seharusnya memberikan output 0 hanya ketika kedua input bernilai 1, sedangkan dalam kondisi lain output akan 1.
- Gerbang logika NOR seharusnya memberikan output 1 hanya ketika kedua input bernilai 0, sedangkan dalam kondisi lain output akan 0.

VII. KESIMPULAN

Pada praktikum di atas bahwa praktikum berhasil dilakukan sesuai teori, hasil pengujian menunjukkan bahwa keluaran rangkaian sesuai dengan tabel kebenaran masing-masing gerbang logika. Keberhasilan ini membuktikan bahwa gerbang logika dasar dapat digunakan untuk membangun sistem digital yang lebih kompleks.

VIII. DAFTAR PUSTAKA

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